

Pregnancy With Preexisting Kidney Disease

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Historically, kidney disease was considered a contraindication to pregnancy, but now many pregnant patients with chronic kidney disease (CKD) have successful outcomes. To achieve this, nephrologists and obstetricians need to be skilled in disease optimization before pregnancy, counseling regarding possible pregnancy outcomes, and antenatal and postpartum management of such patients.

ADVERSE EFFECTS OF CHRONIC KIDNEY DISEASE ON PREGNANCY OUTCOMES

CKD affects up to 6% and 9% of women of reproductive age in high- and low-income countries, respectively, approximately one-third of whom have stages 3 to 5 CKD.¹ Pregnancy rates are lower in women with CKD. In Australia and New Zealand, women with kidney transplants and on dialysis had severalfold fewer births compared with the general population (21.4 vs. 5.8 vs. 61.9 live births/1000 women/year: in transplant, dialyzed, no kidney replacement therapy).²

More recent studies report an increase in live birth rates even in those with more advanced CKD,^{3,4} with considerable improvements in outcomes compared with historic data, likely in part because of enhanced fetal surveillance and neonatal intensive care. The following key prepregnancy factors are the main predictors of pregnancy outcomes:

- Degree of reduction in glomerular filtration rate (GFR)/elevation in serum creatinine
- Presence of coexisting hypertension
- Amount of proteinuria

In most circumstances, these features are more important in predicting outcome than the mother's specific kidney disease etiology.

Traditionally, most women with mild kidney impairment (serum creatinine < 1.5 mg/dL [130 μ mol/L]) and controlled hypertension were thought to have a good pregnancy outcome. However, even CKD stages 1 and 2 are associated with a higher rate of preeclampsia and perinatal mortality, preterm delivery, and small for gestation age rates.^{4,5} Development of superimposed preeclampsia appears higher in women with a preconception creatinine value of less than 1.4 mg/dL (124 μ mol/L) but greater than 0.9 mg/dL (80 μ mol/L) in early pregnancy and, indeed, even in women with normal GFR but dipstick-positive proteinuria and one or more other risk factors for preeclampsia.⁶ Those with serum creatinine greater than 1.5 mg/dL (133 μ mol/L), particularly when accompanied by hypertension and heavy proteinuria, have a higher chance of preterm deliveries and maternal complications, including loss of kidney function. Pregnancy outcomes according to severity of kidney dysfunction are presented in [Boxes 45.1 and 45.2](#).

Severity of Hypertension

Preexisting hypertension is associated with significantly higher risk for adverse pregnancy outcomes than the general population⁷ and

in pregnant women with CKD.⁸ Preexisting hypertension conferred an approximate doubling of risk for severe neonatal outcome (early preterm delivery, requirement for neonatal intensive care, and/or small for gestational age infant) in a prospective Italian cohort study.^{3,9} In a contemporary cohort of women with CKD stages 3 or higher, chronic hypertension was the strongest predictor of preterm delivery before 34 weeks and was associated with a greater decline in maternal kidney function in pregnancy.

Proteinuria

Preexisting proteinuria is another risk factor for poor pregnancy outcome.^{3,10} Women with prepregnancy CKD stages 3 to 5 and a urine protein creatinine ratio greater than 100 mg/mmol (>0.9 mg/mg) before pregnancy or prior to 20 weeks' gestation are more than twice as likely to deliver a baby weighing less than the 10th percentile compared with women with CKD and less proteinuria.

MANAGEMENT COMMON TO ALL PREGNANCIES WITH PREEXISTING KIDNEY DISEASE

The general principles of management of pregnancy in women with CKD are summarized in [Box 45.3](#).

Prepregnancy Counseling

All women with CKD should receive prepregnancy counseling, especially those with CKD stages 3 to 5, but women with CKD 1 or 2 may also have worse pregnancy outcomes than the general population.⁴ Issues that should be covered in counseling are shown in [Box 45.4](#), including relevant information on pregnancy outcomes in CKD shown in [Boxes 45.1 and 45.2](#).

Pregnancy may be unplanned in women with CKD, and detailed discussion regarding maternal and fetal outcomes should be provided in early pregnancy. Furthermore, pregnancy may be the first opportunity for identification of CKD; for example, in a prospective cohort study 16% of women with CKD were diagnosed in pregnancy.¹¹

Fertility Assessment

Fertility is reduced in women with moderate to severe CKD. Some individuals have raised prolactin, luteinizing hormone, and lower anti-müllerian hormone concentrations, suggesting altered pituitary axis response and/or reduced ovarian follicle reserve.^{12,13} With the advent of assisted conception, nephrologists may be approached for advice regarding hormone stimulation therapies and embryo transfer. There is no evidence that hormonal manipulation has additional adverse consequences in women with CKD unless they develop ovarian hyperstimulation syndrome (see [Chapter 44](#)), which may further compromise kidney function. Single embryo transfer is recommended because of the additional maternal and neonatal risk for multiple fetal pregnancies.

BOX 45.1 Maternal Kidney Outcomes According to Prepregnancy CKD Stage**CKD Stage 1 to 2**

- Permanent loss of GFR in <10% of cases
- Greatest risk if GFR is <40 mL/min and proteinuria is >1 g/day
- Major determinant of progressive kidney loss is hypertension
- 40% risk for preeclampsia if baseline proteinuria is >500 mg/day

CKD Stage 3

- Decline or permanent loss of GFR in 30% of cases
 - Increased to 50% if uncontrolled hypertension
 - 10% ESKD soon after pregnancy
 - Accelerated decline in GFR because of pregnancy by 1.7–2.1 years

CKD Stage 4

- Progression to ESKD highly likely during or soon after pregnancy
 - Accelerated decline in GFR because of pregnancy by up to 4.9 years

Note that the Modification of Diet in Renal Disease (MDRD) formula and Chronic Kidney Disease-Epidemiology (CKD-Epi) formula for estimation of glomerular filtration rate are not valid in pregnancy. *CKD*, Chronic kidney disease; *ESKD*, end-stage kidney disease; *GFR*, glomerular filtration rate.

BOX 45.2 Fetal Outcomes According to Maternal Prepregnancy CKD Stage³**Outcomes After Accounting for First-Trimester Miscarriage CKD Stage 1 to 2**

- Live births in most pregnancies
- Up to 50% preterm delivery, up to 18% small for gestational age

CKD Stage 3

- Live births in most pregnancies
- Up to 60% preterm delivery (mainly iatrogenic due to superimposed preeclampsia/fetal growth restriction)

CKD Stage 4 to 5

- Live births in most pregnancies
- 90% preterm delivery, two-thirds small for gestational age, 50% admission to neonatal unit

CKD, Chronic kidney disease.

Note that the Modification of Diet in Renal Disease (MDRD) formula and Chronic Kidney Disease-Epidemiology (CKD-Epi) formula for estimation of glomerular filtration rate are not valid in pregnancy.

BOX 45.3 Principles of Antenatal Care With CKD¹⁰

- Management of hypertension aiming for BP < 135/85 mm Hg
- Aspirin (75–150 mg daily) for all women with CKD
- Regular medication review: Discontinue statins, ACE inhibitors, ARBs
- Correct interpretation of gestational changes in serum creatinine
- Clinical assessment and maintenance of volume homeostasis
- Interpretation and management of proteinuria, including nephrotic syndrome
- Identification of superimposed preeclampsia
- Identification and management of urinary tract infection
- Consideration of the primary kidney disease
- Assessment for fetal well-being and consider if delivery is indicated (see Box 43.5)

ACE, Angiotensin-converting enzyme; *ARBs*, angiotensin receptor blockers; *BP*, blood pressure; *CKD*, chronic kidney disease.

BOX 45.4 Prepregnancy Counseling for Women With Chronic Kidney Disease**Maternal Risks**

- Accelerated decline in glomerular filtration rate, sometimes precipitating dialysis during pregnancy or soon after in women with advanced chronic kidney disease
- Severe maternal hypertension with risk for stroke
- Superimposed preeclampsia with kidney, hepatic, thrombotic, or bleeding and neurologic risks
- Exacerbation of proteinuria, including nephrotic syndrome with risks for thrombosis

Fetal Risks

- Fetal growth restriction or intrauterine fetal death from placental insufficiency
- Preterm delivery, with both short-term and long-term consequences
- Teratogenicity and/or fetotoxicity of some drug therapies for kidney disease
- Inheritance of a kidney disorder

Contraception should be offered to women with CKD of reproductive age who do not wish to conceive, including those on dialysis and with kidney transplants. For the majority, estrogen-containing preparations are contraindicated because of exacerbation of hypertension or thrombotic risk, and progesterone methods including the progesterone-only pill, implant, and intrauterine device are preferred. Women taking immunosuppression have historically been advised against intrauterine devices because of concerns related to failure and pelvic inflammatory disease, although there is no robust evidence to support this recommendation.¹⁴

Urinalysis: Proteinuria

Preexisting proteinuria is often exacerbated during pregnancy because of increased glomerular permeability and plasma blood flow and stopping renin-angiotensin system blockade. However, when there is an increase in protein excretion during pregnancy in women with kidney disease, there are few therapeutic options apart from ensuring blood pressure (BP) control (see later discussion) and managing thrombotic risk. Isolated nonnephrotic proteinuria may develop *de novo* during pregnancy, which usually indicates one of the following scenarios and usually does *not* require antenatal kidney biopsy:

1. Developing preeclampsia (after 20 weeks of gestation).
2. No pregnancy complications occur, proteinuria resolves postpartum (*gestational proteinuria*).
3. Intrinsic glomerular disease has developed either before or during pregnancy and remains postpartum (uncommon). Management of glomerulonephritis (GN) in pregnancy is discussed in more detail later.

Serum albumin falls in most pregnancies as a result of volume expansion and may be less than 30 g/L even in healthy women, thus hypoalbuminemia is not a reliable indicator of nephrotic syndrome. Similarly, peripheral edema and a rise in serum cholesterol are commonly seen in healthy pregnancy. A spot urine protein-to-creatinine ratio (uPCR) greater than 300 mg/mmol (or >3 mg/mg) generally indicates protein excretion is greater than 3 g/day. Diuretics are rarely needed but are indicated for pulmonary edema.

Nephrotic syndrome in pregnancy may create a prothrombotic state and a propensity for intravascular volume contraction (see Chapter 16) that may lead to reduced uteroplacental compromise and worsening maternal kidney function. Treatment requires prophylaxis of thromboembolism with low-molecular-weight heparin, and

maternal kidney function surveillance. Thresholds for commencing heparin are undefined, but most clinicians initiate treatment at higher serum albumin and/or lower proteinuria concentrations than in the nonpregnant state because of the prothrombotic nature of pregnancy. We recommend prophylaxis of venous thromboembolism for all pregnant women with uPCR greater than 300 mg/mmol in pregnancy in the absence of contraindications and where delivery is not anticipated within 12 hours. Lesser degrees of proteinuria (uPCR 100–300 mg/mmol) are considered a risk factor for thromboembolic events, and prophylaxis may be warranted if there are additional risk factors. Low-molecular-weight heparins are safe during pregnancy and do not cross the placenta. Dose adjustment may be indicated for women with CKD. Warfarin is teratogenic and should be avoided during pregnancy.

Hypertension

Many pregnant patients with CKD will not exhibit the usual first-trimester fall in BP, and, in many women, BP increases as the pregnancy progresses. Normal pregnancy is accompanied by significant volume expansion, which does not induce hypertension. However, in CKD, there is often inability to excrete a sodium load with accompanying hypertension. Other potential contributory factors include stimulation of the renin-angiotensin and sympathetic nervous systems; alterations in endothelial factors such as prostacyclin, nitric oxide, and endothelin; and drugs such as calcineurin inhibitors and corticosteroids. Regardless of its cause, persistence of hypertension is an adverse factor in pregnancy outcome and inadequate use of antihypertensives in pregnancy has been associated with poorer pregnancy outcomes, at least in women with kidney transplants.¹⁵ BP often rises after delivery and should be monitored postpartum and treated according to nonpregnant targets, with agents that are safe in lactation.

Hypertension in pregnancy is defined as BP of 140/90 mm Hg or greater (see [Chapter 44](#)). A large randomized controlled trial comparing tight BP control (target diastolic blood pressure [DBP], 85 mm Hg) or less tight control (target DBP, 100 mm Hg) reported no difference in neonatal outcomes, but a reduction in episodes of severe maternal hypertension including requirement for intravenous (IV) antihypertensive treatment and associated liver and platelet abnormalities in those with tighter BP control¹⁶; thus tight blood pressure control is recommended for pregnant women with hypertension.

BP targets for nonpregnant proteinuric women with CKD are lower than pregnancy recommendation. Although there is no direct evidence for optimum BP in pregnant women with CKD, BP less than 130/80 mm Hg may be recommended to avoid prolonged periods with suboptimal control, while avoiding sustained blood pressures less than 110/70 mm Hg. Suitable antihypertensive agents include labetalol, nifedipine, methyl dopa, hydralazine, and prazosin in conventional doses (see [Chapter 44](#)). Choice of first-line agent is based on availability, licensing, and clinician preference and experience. The frequency of dosing may need to be increased because of altered drug metabolism in pregnancy. Diltiazem may have a small benefit in reducing proteinuria, but this has not been addressed specifically among pregnant patients. Diuretics are not recommended during pregnancy because any reduction in maternal plasma volume may have adverse effects on uteroplacental or kidney perfusion but are indicated for the treatment of pulmonary edema or where substantial peripheral edema leads to skin breakdown.

Angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs) cannot be used in pregnancy because of fetotoxicity in the second and third trimesters. It had previously been recommended to cease ACE inhibitors in advance of pregnancy; however, population studies have shown no increased risk compared with women exposed to other antihypertensive agents in the first

trimester.¹⁷ Women with kidney or cardiac indications can therefore continue ACE inhibitors during the unknown length of time it will take to conceive, provided they are counseled to undertake regular pregnancy testing and stop at a positive pregnancy test. There are less data supporting ARB safety, so these agents are usually discontinued in advance of pregnancy. Enalapril or captopril may be used in patients after delivery for those who wish to breastfeed. Aldosterone antagonists should be avoided in pregnancy, and atenolol is associated with fetal growth restriction.

KIDNEY BIOPSY IN PREGNANCY

It is rare to require kidney biopsy in pregnancy except for:

1. De novo onset of nephrotic syndrome or unexplained kidney injury with abnormal urine sediment in the first and early second trimester when the risks of kidney biopsy are lower¹⁸ and when a histologic diagnosis will change management in pregnancy. Nephrotic syndrome after this stage can generally be managed until delivery, with postpartum investigations for those with persistent proteinuria/nephrosis that cannot be attributed to preeclampsia.
2. Where the clinical picture warrants exclusion of acute rejection in a female with a kidney transplant.

Complications of kidney biopsy in pregnancy are more common in the late second and third trimester of pregnancy compared with those in general nephrology practice; a systematic review reported bleeding complications in 7% of biopsies performed in pregnancy compared with 1% performed postpartum.¹⁸

SUPERIMPOSED PREECLAMPSIA

Preeclampsia (see [Chapter 44](#)) is a placental disorder that presents after 20 weeks of gestation and has several predisposing risk factors, including an association with increasing severity of CKD.¹⁹ Superimposed preeclampsia in a female with CKD frequently leads to exaggerated hypertension and proteinuria with risks for nephrotic syndrome, progression of maternal kidney impairment, fetal growth restriction, preterm delivery, and perinatal mortality.

Superimposed preeclampsia is challenging to diagnose in a patient with preexisting hypertension, elevated serum creatinine, and/or proteinuria; however, when accompanied by additional features such as clonus, abnormal liver transaminases, or new-onset thrombocytopenia, superimposed preeclampsia is likely, although few patients with superimposed preeclampsia present with extrarenal manifestations¹¹ and delivery decisions may be required without waiting for other severe systemic features to develop. Proposed criteria for diagnosis of superimposed preeclampsia in patients with CKD are highlighted in [Table 45.1](#).

Placental growth factor (PIGF) testing (reduced in isolation or increased as a ratio to soluble fms-like tyrosine kinase 1 [sFlt-1]) can be used to predict the need for delivery due to preeclampsia (see [Chapter 44](#)). PIGF-based testing has been shown to distinguish preeclampsia without CKD from pregnant patients with CKD without preeclampsia in a small cohort.²⁰ A larger, prospective cohort including patients with chronic hypertension or kidney disease reported comparable diagnostic performance for superimposed preeclampsia and preeclampsia without preexisting disease.¹¹ PIGF-based testing offers a quantitative assessment of placental function and, at gestational concentrations, will not measure maternal endothelial disease caused by CKD. A recent prospective cohort study has shown increased surveillance for need for delivery should take place in patients with CKD and plasma PIGF less than 150 pg/mL after 20 weeks' gestation, compared with a threshold of less than 100 pg/mL in women without CKD. However, this testing

TABLE 45.1 Diagnostic Criteria for Preeclampsia^a

Diagnostic Criteria	Preeclampsia	Preeclampsia in Women With CKD
Essential	>20 weeks' gestation New hypertension: systolic BP $\geq$ 140 mm Hg or diastolic BP $\geq$ 90 mm Hg on two occasions	Hypertension: - In women without chronic hypertension: as for preeclampsia - In women with chronic hypertension: no diagnostic threshold, de novo severe BP (systolic BP > 160 or diastolic BP > 110 mm Hg) or an increase in treatment to maintain BP < 160/110 mm Hg used in research cohorts^{6,8}
Additional	Proteinuria: - uPCR > 30 mg/mmol (or 0.3 μg/mg) - uACR > 8 mg/mmol (or >800 μg/mg) > 300 mg/24 h (not indicated if uPCR available) - Dipstick > 2+ (if other methods unavailable) Serum creatinine: - Serum creatinine >1.0^b to 1.1^c mg/dL (90–100 μmol/L)^d - Doubling of serum creatinine less than 1.1 mg/dL (100 μmol/L)^{c,d}	Proteinuria: - In women with nonproteinuric CKD: as for preeclampsia - In women with proteinuric CKD: no diagnostic threshold, > 100% increase and UPCR $\geq$ 0.3 mg/mg (> 30 mg/mmol) used in research cohorts^{6,8} Serum creatinine: - In women with CKD and preserved excretory function: as for preeclampsia - In women with CKD and abnormal prepregnancy function: no consensus on diagnostic threshold for change in creatinine, > 50% increase within 7 days used in research cohorts⁸

^aDiagnosis requires one essential and one additional clinical feature.^{84–87}

^bModified from Mills KT, Xu Y, Zhang W, et al. A systematic analysis of worldwide population-based data on the global burden of chronic kidney disease in 2010. *Kidney Int.* 2015;88:950–957.

^cModified from Bramham K, Briley AL, Seed PT, et al. Pregnancy outcome in women with chronic kidney disease: A prospective cohort study. *Reprod Sci.* 2011;18:623–630.

^dSevere features.

BP, Blood pressure; CKD, chronic kidney disease; uACR, urinary albumin/creatinine ratio; uPCR, urinary protein/creatinine ratio.

is not needed if increased surveillance is already planned and decisions regarding delivery should not be made on the basis of PIGF results alone. The indications for delivery in patients with preeclampsia are the same in women with or without underlying CKD (Box 45.5). PIGF can be a useful adjunct to interpreting gestational changes in hypertension, proteinuria, and creatinine in patients with CKD.

Limited studies suggest that aspirin is of benefit in reducing the risk of superimposed preeclampsia and perinatal death in patients with underlying kidney disease.²¹ Our practice is to use low-dose aspirin (75–150 mg/day) for all patients with CKD, using data generalized from high-risk cohorts (see Chapter 44).

ASSESSMENT OF FETAL WELL-BEING

Preterm delivery, fetal growth restriction, and stillbirth are the major concerns for women with kidney disease, particularly if superimposed preeclampsia develops. Avoidance of preterm delivery reduced both short-term and lifelong complications, including higher mortality in later life.

Early pregnancy ultrasound between 11 and 14 weeks is advised to estimate accurately the expected date of delivery and to offer trisomy screening, followed by fetal morphology scan at 18 to 20 weeks of gestation to screen for fetal anomalies, assess fetal well-being, and check placental function and position. Blood human chorionic gonadotropin (hCG) levels are often elevated in patients with kidney impairment and may overestimate the risk for fetal abnormalities.²²

Additional ultrasound scans should be offered at 2- to 4-weekly intervals from 28 weeks of gestation in women with CKD to assess fetal growth and amniotic fluid volume, with Doppler studies of umbilical artery blood flow. Absent end-diastolic flow on umbilical artery velocimetry is usually an indication for the need for delivery, depending on gestation. The development of reversed end-diastolic flow indicates a high risk for fetal hypoxia, acidosis, and death.

BOX 45.5 Indications for Delivery in Women With Preeclampsia and Chronic Kidney Disease

- Gestation $\geq$ 37 weeks
- Inability to control blood pressure
- Deteriorating kidney function
- Neurologic abnormalities, such as eclampsia, headaches with accompanying clonus and hyperreflexia, or repeated visual scotomata
- Worsening thrombocytopenia
- Increasing liver transaminase levels
- Failure of fetal growth
- Reversed or absent end-diastolic flow on cardiotocography

The decision to deliver is based on a global clinical assessment rather than on specific thresholds for each parameter.

TIMING OF DELIVERY

In patients with stable CKD and no evidence of fetal compromise, pregnancies should be continued to term and spontaneous labor awaited. The method of delivery, either vaginal or by cesarean delivery, is determined by obstetric issues rather than CKD. Indications for delivery are described in Box 45.5.

The goal is to time delivery such that the risks of prematurity are less than the risks of the pregnancy continuing, which requires multidisciplinary consultation among obstetrician, neonatologist, nephrologist, or obstetric medicine physician and midwife. Decision making regarding the timing of iatrogenic preterm delivery is a balance between the need for fetal maturity and concern regarding maternal health, which for patients with CKD is usually the development of superimposed preeclampsia or worsening kidney function.

Final pregnancy outcomes for patients with CKD are usually successful for both mother and baby, although there may be long-term implications

for childhood development and maternal health. Clinicians can take a positive approach, at all times emphasizing the need for diligence and assessment for potential complications, and aim to relieve some of the stress that accompanies pregnancy for patients with CKD.

COURSE OF CHRONIC KIDNEY DISEASE DURING AND AFTER PREGNANCY

For most women with CKD, the risk for progression of kidney disease during pregnancy is determined by the severity of kidney disease prior to pregnancy, chronic hypertension, and the development of superimposed preeclampsia, rather than the underlying cause of CKD.

Rapid deterioration during a pregnancy is unlikely in patients with mild CKD (prepregnancy CKD stage 1 to 2), other than lupus nephritis, which may flare. However, 50% of patients with moderate baseline kidney impairment (serum creatinine > 1.5 mg/dL [130 μ mol/L]) or worse have a significant rise in serum creatinine in the third trimester or early postpartum, and almost 1 in 5 of these will require kidney replacement within 6 months.²³ In a series of 49 women with stage 3 to 5 CKD before conception whose pregnancy proceeded beyond 20 weeks, GFR was lower after pregnancy than before conception, and this fall was predicted by the combination of preconception GFR less than 40 mL/min/1.73 m² and proteinuria greater than 1 g/day, but not by GFR alone.²⁴ In a more recent study of 179 pregnancies in women with prepregnancy CKD stages 3 to 5 not on dialysis, 46% of women had lost 25% of their prepregnancy eGFR or required kidney replacement therapy by 1 year postpartum. There was a measurable loss of GFR associated with pregnancy, equating to 1.7, 2.1, and 4.9 years of prepregnancy kidney disease for women with prepregnancy CKD stages 3a, 3b, and 4 to 5, respectively. Chronic hypertension was a significant predictor of pregnancy-associated loss of kidney function, and absence of an early fall in maternal serum creatinine was a significant predictor of delivery before 34 weeks' gestation and further loss of maternal kidney function.³

The course of kidney disease postpartum is unpredictable. Some women with stable antenatal kidney function acutely deteriorate postpartum. Surveillance and hypertension control should continue postpartum. ACE inhibitors (captopril or enalapril if breastfeeding) or ARBs (after lactation) can be commenced after delivery for their antiproteinuric effect.²⁵

Norwegian women with biopsy-proven kidney disease had slightly faster progression to a requirement for kidney replacement if their pregnancy required preterm delivery but not if they had preeclampsia.²⁶ A neglected issue is the association between preterm birth, low birth weight, and later-life cardiovascular and kidney disease. This may transmit an increased risk for kidney disease to offspring even if the kidney disease is not hereditary. On balance, it is prudent to consider both preeclampsia and preterm delivery as risk factors for later kidney disease.

MANAGEMENT OF COMMON KIDNEY DISORDERS DURING PREGNANCY

Immunoglobulin A Nephropathy

Pregnancy outcomes for patients with immunoglobulin A (IgA) nephropathy are generally positive, and comparable to other causes of CKD. Long-term follow-up of women with IgA nephropathy diagnosed in childhood showed that pregnancy was complicated by hypertension in half of them and preterm delivery in one-third.²⁷ A Swedish cohort study comparing pregnancy outcomes in women with and without IgA nephropathy described an increased risk of preterm birth

(adjusted odds ratio [OR], 2.69), preeclampsia (adjusted OR, 4.29), and small for gestational age (adjusted OR, 1.84), but no reduction in live birth²⁸ rates, although stratification according to CKD stage was not possible. Pregnancy in IgA nephropathy did not accelerate kidney dysfunction in those with CKD stage 1 or 2,^{29–31} but proteinuria may increase in the third trimester.³² However, there is limited evidence that women with proteinuria before pregnancy³³ and those with preterm delivery, low birth weight, and preeclampsia³⁴ have a more rapid decline in function postpartum.

Diabetic Kidney Disease

In the United Kingdom, more pregnant women have type 2 than type 1 diabetes, and 8% of pregnant women with type 1 diabetes and 5% with type 2 diabetes have diabetic kidney disease (DKD).³⁵ Diabetes per se increases risks for congenital abnormality, preterm birth, cesarean section, and perinatal mortality. The presence of overt nephropathy more than doubles the risk for fetal death after 20 weeks. There is an added risk for congenital abnormalities if blood sugar was not adequately controlled at the time of conception, which is not uncommon. For example, the UK National Audit reported that 7 out of 8 women were not well prepared for pregnancy according to national guidelines.³⁵ A large cohort study indicated that the presence of DKD is an independent risk factor for congenital abnormalities.³⁶

The Diabetes Control and Complications Trial and EURODIAB trial both concluded that pregnancy does not increase progression to albuminuria. It is likely that women with DKD have worse pregnancy outcomes than women with other causes of CKD and comparable kidney function. For example, a retrospective cohort study of women with type 1 diabetes and DKD (median prepregnancy serum creatinine 0.8 mg/dL [70 μ mol/L]) reported high complication rates (e.g., preeclampsia 42%, preterm before 34 weeks 21%, neonatal care admission 49%).³⁷ In a recent cohort of women with moderate and advanced CKD, none of the 11 women with DKD had a normal pregnancy course.³

The outcome of DKD during pregnancy depends on the usual factors of severity of kidney impairment and hypertension. Specialized meticulous control of blood glucose and BP during pregnancy is required. One study showed that failure to achieve mean arterial pressure (MAP) less than 100 mmHg was associated with increased risk for early delivery, even after adjusting for glucose control.³⁸ Risk for permanent decline in kidney function for women with more severe DKD has not been formally evaluated. Ideally, ACE inhibitors (enalapril or captopril if breastfeeding) should be introduced soon after delivery for prevention of progression of DKD.

Having gestational diabetes mellitus (GDM) without developing overt diabetes after the pregnancy is still associated with approximately 10% risk for microalbuminuria by age 50, an intermediate risk between that of nondiabetic and diabetic women.³⁹ This means that women with a history of GDM, not just those with type 1 or 2 diabetes, should be followed after pregnancy because of their risk for later developing diabetes.

Lupus Nephritis

Women with active disease at conception are more likely to have active lupus nephritis during pregnancy, which is then associated with high fetal risk, with approximately 1 in 5 pregnancies ending in miscarriage, stillbirth, or neonatal death.⁴⁰ Ideally, women with lupus should be in remission for at least 6 months before conception on medications known to be safe in pregnancy. Prepregnancy assessment may include cessation of renin angiotensin blockade to allow assessment of proteinuria, with repeat kidney biopsy before attempts to conceive if active disease is suspected.

In addition to low-dose aspirin to reduce the risk of superimposed preeclampsia, all women with previous lupus nephritis should receive hydroxychloroquine.⁴¹ Hydroxychloroquine in pregnancy is associated with reduced risks of lupus flare and fetal growth restriction.^{42–44}

Lupus nephritis flares are described in 30% of women during pregnancy and in 15% of women immediately postpartum,⁴⁵ but a prophylactic postpartum increase in corticosteroid dosing is not indicated. Although rare, the major cause of maternal death in lupus nephritis is sepsis, so immunosuppression should be maintained at the lowest possible dose to control disease.⁴⁶ New onset or a flare of lupus nephritis, evidenced by increasing proteinuria, active urine sediment, and rising serum creatinine, is a major concern. Distinguishing a flare of lupus nephritis from preeclampsia can be challenging because the phenotype of both conditions includes hypertension, proteinuria, kidney injury, and thrombocytopenia. A kidney biopsy can be arranged in the first and early second trimester when bleeding risks are lower, and only if a histologic diagnosis will change management in pregnancy. Serological markers of disease, including complement and double-stranded DNA antibody concentrations, may also be clinically helpful.

Lupus nephritis can be treated with corticosteroids and azathioprine in pregnancy. Tacrolimus is an alternative agent that has been used with increasing success in women with active lupus nephritis in pregnancy.⁴⁷ Cyclophosphamide and mycophenolate mofetil (MMF) are contraindicated in pregnancy because of teratogenicity and should be ceased at least 6 weeks before conception. MMF is associated with microtia (underdeveloped external ear), auditory canal atresia, cleft lip and palate, and micrognathia. Although cyclophosphamide has been used successfully in a few cases of lupus nephritis in later pregnancy after the period of fetal organogenesis, its use is not routinely recommended. Rituximab is not teratogenic but crosses the placenta and causes B-cell depletion in the fetus as in the mother⁴⁸ and is therefore not recommended for use in pregnancy, where other treatment options exist.⁴⁹ The use of belimumab in pregnancy is described in isolated case reports, with insufficient data to confirm safety.⁵⁰ There are no pregnancy-specific contraindications to the use of IV immunoglobulin and plasma exchange in pregnancy if clinically indicated.

Prior histologic class of lupus nephritis has no influence on pregnancy outcome. Predictors of poor pregnancy outcome include baseline creatinine greater than 0.9 mg/dL (79 μ mol/L), proteinuria greater than 0.5 g/24 hours, antiphospholipid syndrome, hypertension, non-White ethnicity, and maternal disease flare.⁵¹ A prospective cohort study of women with lupus, including those with mild nephritis, recently reported that pregnancy outcomes were favorable for women without adverse risk factors (including lupus anticoagulant, antihypertensive use, active disease, or low platelets), with 92% having uncomplicated pregnancies. However, when counseling about future pregnancies, women should be advised that even if their lupus nephritis has been well treated, they appear to have a greater risk for maternal complications and lupus flare in their next pregnancy than do women with systemic lupus erythematosus (SLE) who have never had nephritis; this is only partly explained by increased activity of the SLE at conception. Fortunately, fetal outcomes are not affected adversely in this group and postpartum GFR appeared unaffected.⁵² This means that such patients need more surveillance during pregnancy than usual, to detect a kidney flare or preeclampsia, their two biggest maternal risks.

Women with anti-Ro and anti-La antibodies should be counseled regarding the risk for fetal heart block and neonatal cutaneous lupus, respectively. Hydroxychloroquine may reduce the risk for recurrent fetal heart block in patients with a previously affected infant, and this is indicated in pregnancy in women with lupus nephritis.⁵³ There is debate as to the optimum monitoring for congenital heart block with

variable recommendations for fetal heart rate monitoring and echocardiography in the second trimester. Even if surveillance detects early-stage fetal heart block, the optimum treatment is unclear.

Reflux Nephropathy

Pregnancy outcomes in women with reflux nephropathy depend predominantly on preexisting kidney function and BP control as for other kidney diseases. An analysis of contemporary data (2000–2016) including 434 women with 879 pregnancies showed an increased risk of pregnancy-related hypertension (OR, 5.55) and preeclampsia (OR, 6.04), but no difference in preterm delivery compared with unmatched control pregnancies. A higher incidence of stillbirth has been reported, which warrants further analysis.

Women with reflux are more predisposed to urinary tract infection (UTI) throughout pregnancy, as are women who have had surgically corrected reflux in childhood. Around 20% of women with reflux nephropathy will develop UTI in pregnancy, with a 6% incidence of pyelonephritis in historical studies. Urine culture should be performed throughout pregnancy in those with underlying reflux nephropathy and treatment offered for confirmed bacteriuria. In the absence of evidence of harm, antibiotic prophylaxis is recommended to those with reflux nephropathy after a single confirmed infection, informed by urine culture, antimicrobial sensitivities, and patient preference.⁴⁹ Management of UTI in pregnancy is further discussed in Chapter 44.

Vesicoureteric reflux may be a familial disorder, estimated to affect 20% of infants with a positive parental family history, compared with a 1% to 2% frequency in the general population.⁵⁴ However, not all vesicoureteric reflux results in kidney parenchymal damage, and 80% of mild cases resolve in childhood (see Chapter 62).⁵⁵ Data are inadequate to define an evidence-based screening strategy for the offspring of women with reflux nephropathy. Expert consensus, based on observational data, is that infants of mothers with urinary tract abnormalities who had normal urinary tracts on antenatal ultrasound scans do not need further follow-up unless features of UTI are identified in childhood.⁴⁹

Inherited Kidney Disorders

Although chronic hypertension is a risk factor for adverse pregnancy outcomes in autosomal dominant polycystic kidney disease (ADPKD),⁵⁶ normotensive women with ADPKD have comparable pregnancy outcomes to those of unaffected family members. A study of 54 women with ADPKD compared with 92 women with simple cysts showed increased rates of hypertension and preeclampsia but no difference in neonatal outcome.⁵⁷ Case reports suggest that there is no change in kidney volume in ADPKD because of pregnancy and uterine growth is not compromised.

Very few pregnancies have been described in women with Alport syndrome, but substantial gestational proteinuria with clinical nephrosis is recognized.⁵⁸

Inherited kidney disorders are likely to have been diagnosed before pregnancy, and specific implications for the offspring should be discussed. For single-gene disorders, preimplantation diagnosis may be available, and referral for specialist counseling is recommended.

HEMODIALYSIS IN PREGNANCY

Significant improvements in outcomes of pregnant patients requiring dialysis during pregnancy have occurred over the past two decades, with fetal survival rising from less than 50% to almost 85%.⁵⁹ Pregnancy rates have also risen, with a 71% increase in delivery rate among women of reproductive age undergoing hemodialysis reported in the United States⁶⁰ from 2002 to 2015. Pregnancy rates in non-White women

treated with maintenance dialysis are also reported to be higher.⁶¹ Improved outcomes have been associated with more intensive hemodialysis regimens, and advances in neonatal care have enabled survival for premature and growth-restricted infants. A Canadian cohort study reported significant improvement in birth weight and length of gestation in women who received prolonged dialysis (43 ± 6 hr/wk) compared with those with a more modest increment (17 ± 5 hr/wk).⁶² A meta-analysis of 574 pregnancies in 543 patients reported that fewer dialysis hours per week were strongly associated with preterm delivery before 37 weeks.⁶³ Recommendations for managing hemodialysis during pregnancy are listed in Box 45.6.

Women of childbearing age undergoing dialysis have low rates of conceiving,⁵⁹ but increments in dialysis dose, such as nocturnal home hemodialysis, appear to be associated with increased fertility.⁶² Women of childbearing age on maintenance dialysis who do not wish to conceive should therefore have access to safe and effective contraception (progesterone-only pill, progesterone implant, intrauterine device).

Initiating Dialysis for Progressive Chronic Kidney Disease

Dialysis should be commenced in pregnancy for the standard indications of hyperkalemia, acidosis, uremic symptoms and/or fluid overload. However, it is more common to initiate dialysis in pregnancy because of potential fetotoxicity of urea. The optimum serum urea concentration at which dialysis offers benefit in pregnancy is unknown and practice varies from routine commencement of dialysis at maternal blood urea nitrogen (BUN) 48 mg/dL⁶⁴ (or urea > 17 mmol/L) to dialysis only when BUN is consistently greater than 56 mg/dL (or urea > 20 mmol/L). We recommend that in progressive kidney disease, BUN greater than 42 mg/dL (or serum urea > 15 mmol/L) should initiate conversations about the risks, benefits, and logistics of hemodialysis in pregnancy, weighed with the risks of preterm delivery before dialysis initiation if the gestation is approaching or exceeds 34 weeks.⁴⁹ Dialysis initiated during pregnancy is probably associated with greater likelihood of successful pregnancy than for those on maintenance dialysis, perhaps because of the benefits of residual kidney function,⁶⁵ although this is not a consistent finding.⁶²

Dialysis Regimens in Pregnancy

Differences in residual kidney function means that intensification of dialysis may not show the same benefit for women initiating dialysis in pregnancy as it does for women established on hemodialysis before pregnancy. Meta-analysis data demonstrating improved outcomes with intensification of dialysis do not include women starting dialysis after 20 weeks' gestation.⁶³ In a study of intensive dialysis in pregnancy, a very small subgroup analysis of women starting hemodialysis after conception showed no significant difference in gestational age at delivery between those receiving 33 ± 6 hours per week ($n = 4$) compared with 15 ± 4 hours per week ($n = 13$).⁶²

A graded increase in dialysis guided by residual kidney function and biochemical parameters has been described with initial weekly hours of hemodialysis determined according to residual urine output and time on dialysis before pregnancy, with dialysis provision then adjusted according to midweek predialysis serum urea, BP, weight gain, and polyhydramnios. In this cohort, linear regression identified a midweek predialysis BUN 35 mg/dL (or serum urea 12.5 mmol/L) as discriminatory in determining successful pregnancy outcome.⁶⁶ We therefore recommend initiating hemodialysis at 2 to 3 hours, 3 times per week in women newly starting dialysis in pregnancy for a fetal rather than maternal indication, and adjusting provision to achieve midweek predialysis urea less than 12.5 mmol/L (BUN 35 mg/dL).

BOX 45.6 Managing Hemodialysis During Pregnancy⁴⁹

Prepregnancy

Discuss risks of pregnancy (miscarriage, fetal death, fetal growth restriction, preterm delivery, preeclampsia)

Ensure all medications are safe in pregnancy

Aspirin: 75–150 mg daily

Folic acid: 5 mg daily

During Pregnancy

Dialysis

- Aim for midweek predialysis BUN <35 mg/dL (serum urea 12.5 mmol/L)
- Heparin requirement may increase because of hypercoagulability of pregnancy

Anemia

- Intravenous iron to maintain iron stores
- Dose ESA to achieve hemoglobin 10–11 g/dL

Bicarbonate

- Adjust dialysate bicarbonate to achieve normal serum bicarbonate for pregnancy (18–22 mmol/L)

Nutrition

- Dietician advice to ensure adequate protein and nutrient intake.
- Supplement oral or dialysate phosphate to maintain postdialysis serum phosphate in normal range.

Calcium

- Maintain normal serum calcium with vitamin D/vitamin D analogues and appropriate dialysate calcium.

Phosphate

- Supplement oral or dialysate phosphate to maintain postdialysis serum phosphate in normal range.

After Pregnancy

Return to usual dialysis schedule.

Readjust dry weight and antihypertensives weekly for 6–12 weeks.

BUN, Blood urea nitrogen; ESA, erythropoiesis-stimulating agents.

In addition to augmentation of dialysis dose as practically possible, ensuring the aforementioned biochemical goals are achieved, and weekly surveillance, the following also should be considered:

- Control of maternal BP, generally to less than 110 to 140/70 to 90 mm Hg. This is difficult to achieve in many cases. Volume expansion is often evident in women on maintenance hemodialysis who become pregnant, as evidenced by anemia and fall in serum albumin. There are no data supporting assessment of volume status in pregnant patients on dialysis using ultrasound measurement of inferior vena cava diameter or bioimpedance.
- IV iron and erythropoiesis-stimulating agents aiming for a hemoglobin level of 10 to 11 g/dL. No adverse consequences in pregnancy have been reported with either IV iron or erythropoiesis-stimulating agents, although there is a paucity of data on IV iron preparation use in the first trimester and a theoretic risk for exacerbation of hypertension with the latter agents. Transfusion should be avoided to minimize sensitization but may be required if hemoglobin levels are low, especially before delivery.
- Phosphate monitoring and supplementary phosphate in the dialysate if required.
- Fetal monitoring with at least 2- to 4-weekly ultrasound scanning from the time of fetal viability, around 24 weeks of gestation.

Peritoneal Dialysis and Pregnancy

There is little information concerning specific requirements of women on peritoneal dialysis (PD) during pregnancy. A systematic review and meta-analysis compared pregnancy outcomes in women on hemodialysis (HD) with PD and reported a higher incidence of infants who are small for gestational age in those receiving PD.⁶³ Pregnancy rates in Medicare-insured women with ESKD (2000–2015) were only 2.8% (0.5–1.4 deliveries per 1000 patient-years), and, unlike the number of pregnancies in women receiving hemodialysis or kidney transplants, have not increased over the last 2 decades.⁶⁰ A major risk is that PD peritonitis may provoke premature labor or preterm rupture of membranes. Although some women have had successful outcomes remaining on PD throughout pregnancy, we recommend switching to HD to allow augmentation of dialysis dose in a controlled manner. This recommendation is predicated on the greater available evidence for HD in pregnancy and may change when more data emerge about PD in pregnancy.

KIDNEY TRANSPLANTATION AND PREGNANCY

Successful kidney transplantation is an excellent way of restoring fertility in women with ESKD. It is surprising therefore that the pregnancy rate among female transplant recipients more than halved between 1990 and 2003, which is unexplained by a change in the age of women being transplanted but may be related to the introduction of MMF, which is teratogenic.⁶⁷ Women with transplants appear less likely to conceive if ESKD is caused by diabetes compared with women without diabetes and overall pregnancy rates among kidney transplant recipients are only one-third of the general population.²

Timing of Pregnancy

Women are recommended to wait at least 12 months after a successful kidney transplant before embarking on pregnancy to ensure stable transplant function and maintenance immunosuppression,⁴⁹ although evidence is conflicting. A meta-analysis of pregnancy outcomes in women with kidney transplants reported that cesarean section rate was higher and live birth rate lower in those who conceived in the interval of 2 to 3 years than greater than 3 years and less than 2 years (68% vs. 75% vs. 74%, and 73% vs. 65% vs. 42%, respectively). Maternal complications of preeclampsia were higher in the 2- to 3-year interval, and the greater than 3-year interval compared to the less than 2-year interval (24% vs. 23% vs. 13%). Spontaneous abortion rates were highest in the greater than 3-year interval followed by the 2- to 3-year interval, and the less than 2-year interval (16% vs. 14% vs. 10%).⁶⁸ However, in a study in the United States, rates of preeclampsia, gestational diabetes, cesarean section, and preterm delivery also were higher than for the general population in those who conceived within 2 years.⁶⁹ One to 2 years after transplantation seems a practical time for ensuring clinical stability and having optimal BP control, absence of cytomegalovirus (CMV) viremia, and stable immunosuppression.

Immunosuppression in Pregnancy

There are usually opportunities to ensure that conception is carefully planned, immunosuppression and vaccination status optimized, and folic acid commenced, but, unfortunately, many pregnancies in women with kidney transplants are still unplanned. Discussion regarding restoration of fertility with transplantation and contraception should take place before transplantation.⁷⁰ Immunosuppressive drugs considered safe in pregnancy include prednisolone, azathioprine, cyclosporine, and tacrolimus. Tacrolimus has been associated with neonatal hyperkalemia, but overall data support its safety. MMF is associated with teratogenic effects and should be avoided in pregnancy,

as should sirolimus and everolimus. We suggest ceasing such drugs 3 to 6 months before pregnancy to ensure clinical stability on an alternative agent (usually azathioprine).

Gestational diabetes occurs in 3% to 12% of pregnancies of patients with kidney transplants but no more frequently in tacrolimus-treated patients than in those receiving cyclosporine. Screening for gestational diabetes is recommended for all transplant recipients at 12 and 28 weeks.

Dosing of tacrolimus in pregnancy remains problematic until assays for free tacrolimus become readily available. During pregnancy, tacrolimus shows increased metabolism but altered protein binding.⁷¹ Our practice is to change dose from prepregnancy doses when trough levels are less than normal, with increments targeting the lower end of the prepregnancy range, but with return to prepregnancy doses immediately after delivery, with ongoing close monitoring.

Azathioprine, tacrolimus, cyclosporine, and prednisone appear safe to use for patients who are breastfeeding, but the decision to breast-feed remains an individual one. Patients can be informed that breast-feeding may have considerable advantages, particularly in preterm and growth-restricted babies.

Pregnancy Outcomes

A favorable view of pregnancy in patients who have undergone successful kidney transplantation was supported by observations in over 3000 pregnancies from 2000 women mostly receiving azathioprine and prednisone.⁷² About 15% of these pregnancies miscarried spontaneously, and of those going past the first trimester, pregnancy was successful in over 90% of cases, provided that hypertension or a decline in GFR had not occurred before 28 weeks of gestation, in which case successful pregnancy outcome was reduced to about 70%. Patients with preconception serum creatinine of 1.4 mg/dL (125 μ mol/L) had 96% pregnancy success, whereas those with higher serum creatinine had a 75% success rate. In keeping with the data for all women with CKD, long-term decline in kidney function occurred significantly more often (27%) in those with preconception serum creatinine greater than 1.4 mg/dL (125 μ mol/L).

A recent systematic review of 78 studies reporting 6712 pregnancy outcomes in 4174 kidney transplant recipients reported the following pregnancy outcomes: live birth rate of 72.9%, miscarriage rate of 15.4%, termination of pregnancy in 12.4%, stillbirth rate of 5.1%, and ectopic pregnancies in 2.4%. Maternal complications included preeclampsia in 21.5%, cesarean section in 62.6%, gestational diabetes in 5.7% (US mean, 9.2%), and gestational hypertension in 24.1%. Neonatal outcomes included preterm birth (<37 weeks) in 43.1% with mean gestational age at delivery of 34.9 weeks; neonatal mortality was 3.8%.⁶⁸

Graft and patient survival are similar in those with and without any pregnancy over a follow-up as long as 15 to 20 years, based on 577 pregnancies in the Australian and New Zealand Data registry, most of whom had GN or reflux nephropathy as their primary diagnosis.⁷³ A more recent meta-analysis including 2453 recipients with pregnancies confirmed stable serum creatinine postpartum, with no differences in rates of graft loss compared with nulliparous controls,⁷⁴ and a Norwegian population study has confirmed reduced rates of graft loss and death in women with kidney transplants after pregnancies compared with nonpregnant women despite adjustment for possible confounders.⁷⁵

More recently, small studies have also reported longer-term outcomes for children of women with kidney transplants with normal development at 12 and 24 months in 94%, despite low birth weights and prematurity⁷⁶ (Box 45.7).

BOX 45.7 Maternal and Fetal Outcomes in Kidney Transplant Pregnancies

Maternal

- Decline in glomerular filtration if preconception serum creatinine >1.4 mg/dL (125 μmol/L); rate of decline not usually different from female transplant recipients who have not given birth
- Graft loss in 5% to 10% at 2 years postpartum
- Hypertension in about two-thirds
- Rejection reported in 2% to 9% (risk not thought to be increased by pregnancy)
- Infection in 20% to 35%
- Gestational diabetes mellitus in 6%

Fetal

- Overall live birth rate: 75% to 80%
- Spontaneous miscarriage: 15%
- Preterm delivery: 45%
- Growth restriction: 50%
- Birth defects in about 5%, comparable to the general population
- Long-term consequences of preterm delivery, although most infants have normal postnatal growth and development

Transplant Rejection

Acute transplant rejection is uncommon—reported in fewer than 1 in 20 cases in older series,⁶⁹ but a recently systematic review reported the overall acute rejection rate during pregnancy among 822 kidney transplant recipients to be 9.4% (95% confidence interval [CI], 6.4–13.7), which is comparable to the U.S. mean of 9.1%. The clinical presentation of acute graft dysfunction is not different in pregnancy but differentiation from preeclampsia may be challenging. A recent case control study using Transplant International Pregnancy Registry data identified that acute rejection is associated with higher creatinine concentrations, whereas patients with preeclampsia had much higher levels of proteinuria.⁷⁷ Kidney biopsy is required to confirm the diagnosis. Treatment is limited to IV methylprednisolone and increments in oral immunosuppression because safety of more potent agents (e.g., antithymocyte globulin) is unproven in pregnancy and likely to cause harm to the developing fetus. Delivery may need to be considered at more advanced gestations to facilitate therapies with unknown fetal safety.

Infection in Pregnant Transplant Recipients

There is an increased risk for infection (20%–35%), particularly UTI but also CMV infection, with attendant maternal and fetal risks. The consequences of any infection can include preterm labor and preterm rupture of membranes. CMV can cause congenital malformations and fetal demise, but delay in conception to after at least 1 year posttransplantation makes new infection or reactivation unlikely. Valganciclovir prophylaxis is not advised because of teratogenicity in animals, but there are case reports of successful use in women with severe CMV infection in pregnancy.

Recommendations for management of pregnancy in patients with a kidney transplant are summarized in [Box 45.8](#).

Male Transplant Recipients

Outcomes of pregnancies fathered by male transplant recipients showed mean gestational age and mean birth weight similar to those of the general population. There have been recent concerns about teratogenicity of MMF in the offspring of fathers with kidney transplants,

BOX 45.8 Management of Kidney Transplant Patients During Pregnancy

Prepregnancy

- Stable graft function at least 1 year after transplantation
- Discuss risks with transplant recipient and partner
- Best pregnancy outcome will occur if:
 - Prepregnancy serum CKD stage 1 to 2
 - Proteinuria < 1 g/day
 - Blood pressure < 130/80 mm Hg
- Replace all medications that are not safe in pregnancy
- Eradicate UTI before pregnancy; consider prophylactic antibiotics if there has been recurrent UTI in the past year or a single confirmed UTI in pregnancy
- Stable cyclosporine or tacrolimus blood levels
- Test for and control diabetes

During Pregnancy

- Aspirin (75–150 mg daily) for all women with kidney transplant unless contraindicated.
- Assess fetal growth by ultrasound at least every 4 weeks from 28 weeks of gestation.
- Consider increase in CNI dose during pregnancy if drug level is significantly lower than the nonpregnant target range.
- Screen for gestational diabetes according to local protocol with additional early screening if taking steroids or calcineurin inhibitor.
- At each visit, assess BP (goal 110–135/70–85 mm Hg), proteinuria (dipstick, then uPCR or uACR if dipstick result is positive), urine culture, electrolytes, creatinine, full blood count, predose CNI level, and fetal growth.
- Reassess at each visit whether there is an impending indication for delivery (see [Box 45.5](#)).

During Delivery

- Vaginal delivery is generally possible, and indications for caesarean delivery are usually obstetric.
- Prophylactic antibiotics are not required.

Postpartum

- Control blood pressure.
- Monitor CNI and serum creatinine levels.
- CNI drugs, azathioprine, and corticosteroids are safe in breastfeeding.
- Higher level of clinical surveillance for the first 3 months.

BP, Blood pressure; CKD, chronic kidney disease; CNI, calcineurin inhibitor; uACR, urinary albumin/creatinine ratio; uPCR, urine protein/creatinine ratio; UTI, urinary tract infection.

and the manufacturers have advised switching to a different agent before conception. However, these recommendations are largely theoretical, and should be balanced against the risk for inadequate immunosuppression for a prolonged period in males with higher risk transplants. A recent French survey reported no increased risk of congenital malformations in offspring of males with kidney transplants compared with the general population.⁷⁸ Similarly, the ANZDATA registry recently reported no increase in congenital abnormalities in offspring of fathers taking MMF or derivatives.⁷⁹

PREGNANCY IN THE KIDNEY DONOR

It is usually stated that being a kidney donor does not adversely affect future pregnancy outcomes in terms of fetal birth weight, stillbirth, or prematurity, although recent studies suggest there may be an increased risk for preeclampsia in pregnancy after organ donation, and some

reports suggest that fetal outcomes also may be slightly worse.^{80,81} A large Canadian study also reported a higher incidence of preeclampsia and gestational hypertension in women after donation compared with controls but there were no differences in neonatal outcomes, including gestation at delivery or birth weight, and thus placental insufficiency is unlikely to be severe.⁸²

More recently, a U.S. case-control study also demonstrated an increased risk of preeclampsia in primiparous kidney donors compared with age- and race-matched primiparous controls, but this was only significant for women up to 30 years old,⁸³ and in keeping with preexisting studies there were no differences in neonatal outcomes. These studies have some limitations in their design, and pregnancy should not be discouraged in women who have been kidney donors; rather, the key message is that all such women should be treated as at-risk pregnancies and have a higher number of clinical reviews, focusing on maternal BP and urinalysis and fetal growth, than in low-risk pregnancies. Aspirin prophylaxis may be considered, especially if women have additional risk factors for preeclampsia.

SUMMARY

A summary of factors to be considered in managing a pregnant patient with kidney disease is shown in **Box 45.9**. Attention to these issues from preconception to postpartum can result in good pregnancy outcomes with preservation of maternal health. These patients and, in many cases, their offspring require lifelong assessment for progressive kidney and/or cardiovascular disease.

BOX 45.9 Managing Women With Preexisting Kidney Disease During Pregnancy

- Women with chronic kidney disease (CKD) should be managed by a team made up of the obstetrician, nephrologist or expert physician, and experienced midwife or specialty nurse, preferably in a high-risk pregnancy clinic.
- Main determinants of pregnancy outcome are prepregnancy GFR, proteinuria, blood pressure (BP), medication, and previous obstetric history. These should be the focus in counseling.
- Kidney function should be monitored using serum creatinine in pregnancy.
- Proteinuria is quantified using spot uPCR or uACR.
- Low-dose aspirin should be given to reduce the risk for preeclampsia.
- Low-molecular-weight heparin for VTE prophylaxis if uPCR > 300 (uACR > 200). Assess VTE risk if uPCR > 100 (uACR > 30); may need dose adjustment for kidney function.
- Primary issues during pregnancy are BP control, surveillance for preeclampsia, and regular assessment of fetal well-being.
- Fetal assessment by ultrasound
 - At 20 weeks: Uterine artery pulsatility index
 - Monthly from 28 weeks: fetal growth, blood flow, amniotic fluid index
- Plans for obstetric, nephrology, and neonatal review determined at postpartum discharge. Increased nephrology surveillance recommended for 3 months. Pre-pregnancy counseling recommended in advance of a future pregnancy.

GFR, Glomerular filtration rate; *uACR*, urinary albumin/creatinine ratio; *uPCR*, urine protein-to-creatinine ratio; *VTE*, venous thromboembolism.

SELF-ASSESSMENT QUESTIONS

1. A 32-year-old woman with ADPKD wishes to become pregnant. Her GFR is 110 mL/min/1.73 m². Her BP is 142/89 mm Hg, and she has no proteinuria. She is taking amlodipine 5 mg once daily. Which of the following is correct advice about a future pregnancy?
 - A. Her pregnancy will be low risk.
 - B. She should be offered preimplantation genetic testing.
 - C. She should take aspirin daily in advance of pregnancy.
 - D. She should stop amlodipine when she becomes pregnant.
2. A 39-year-old woman with ESKD secondary to lupus nephritis received a kidney transplant from her partner 9 months ago and wishes to conceive. She has had no episodes of rejection. Her GFR is 84 mL/min/1.73 m², and her BP is 124/77 mm Hg. She is currently taking prednisolone, tacrolimus, MMF, and aspirin. What would you recommend to achieve the best outcome?
 - A. She should not conceive until at least 24 months after transplantation.
 - B. Breastfeeding is not recommended with immunosuppressant medication.
 - C. MMF should be stopped.
 - D. Anti-Ro and La antibody should be assessed to inform fetal monitoring.
3. A 28-year-old woman with IgA nephropathy is at 32 weeks of gestation. Her prepregnancy GFR is 45 mL/min/1.73 m² and uPCR is 120 mg/mmol. She presents with headache and peripheral edema; her BP is 166/100 mm Hg and uPCR is 245 mg/mmol. Her serum creatinine has increased from 70 μmol/L 4 weeks ago to 84 μmol/L. What is the best approach to management?
 - A. Admit for BP control and assess by fetal artery Doppler to determine timing of delivery.
 - B. Admit for BP control and urgent cesarean section delivery to preserve kidney function.
 - C. Give IV magnesium sulfate to prevent eclampsia, then plan delivery within 24 hours.
 - D. Start aspirin 75 mg daily and labetalol 200 mg three times daily with escalation to IV antihypertensives if no response.

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